

PITTSBURG-BUTLER, PA, USA

WMO: 725124

Lat: 40.777N

Lon: 79.950W

Elev: 380

StdP: 96.84

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.4	-21.3	0.6	-14.8	-18.3	0.8	-11.9	9.9	-0.2	8.7	-1.4	2.2	260	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	31.2	22.4	29.2	21.6	28.0	20.8	23.8	28.8	23.0	27.6	22.1	26.4	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.9	26.7	21.4	16.8	25.4	21.0	16.4	25.0	73.2	29.0	70.0	27.8	66.5	26.0	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.1	6.0	-19.9	32.8	4.1	1.7	-22.8	34.0	-25.2	35.0	-27.5	35.9	-30.5	37.1		
(5)			-20.1	25.2	3.7	1.0	-22.8	25.9	-25.0	26.5	-27.1	27.1	-29.8	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-2.7	-1.4	3.3	9.8	15.1	19.7	21.8	20.9	17.2	11.0	5.2	0.3	(6)
(7)		DBStd	9.92	6.81	6.41	6.20	5.30	4.56	3.45	2.82	2.84	3.86	4.64	5.18	5.56	(7)
(8)		HDD10.0	1531	398	324	227	67	8	0	0	0	1	43	161	303	(8)
(9)		HDD18.3	3345	653	554	468	259	121	26	5	9	66	231	395	558	(9)
(10)		CDD10.0	1559	3	3	17	62	167	292	365	338	217	75	16	3	(10)
(11)		CDD18.3	331	0	0	1	4	22	67	112	89	32	5	0	0	(11)
(12)		CDH23.3	2649	0	0	4	43	214	549	895	658	252	32	0	0	(12)
(13)		CDH26.7	691	0	0	0	5	41	144	263	172	61	5	0	0	(13)
(14)	Wind	WSAvg	2.5	3.2	3.0	3.0	3.0	2.5	2.1	1.8	1.7	1.9	2.4	2.7	3.0	(14)
(15)	Precipitation	PrecAvg	1037	86	65	84	89	94	104	105	93	93	72	81	83	(15)
(16)		PrecMax	1390	180	137	140	154	146	169	243	191	225	161	138	182	(16)
(17)		PrecMin	749	55	31	42	45	35	28	35	40	43	20	22	39	(17)
(18)		PrecStd	144	31	23	29	26	32	39	49	40	46	29	32	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	17.8	23.0	27.4	30.0	32.1	32.7	32.4	31.2	27.1	21.2	17.2	(19)
(20)			MCWB	13.3	11.6	14.8	16.9	20.9	22.7	23.7	22.2	21.6	19.9	14.9	13.3	(20)
(21)		2%	DB	13.3	14.1	19.8	24.0	27.7	30.0	31.2	30.3	28.2	23.0	17.9	13.9	(21)
(22)			MCWB	11.0	9.9	12.5	14.9	19.1	21.8	22.9	21.9	20.6	16.7	12.8	10.9	(22)
(23)		5%	DB	11.0	11.3	17.0	22.1	26.2	28.0	29.1	28.6	26.3	21.3	16.2	12.0	(23)
(24)			MCWB	8.9	7.5	11.0	14.1	18.2	21.0	22.1	21.3	19.6	15.9	11.5	9.6	(24)
(25)		10%	DB	7.3	7.7	13.0	18.9	23.6	27.0	27.8	27.2	24.0	18.8	13.8	8.7	(25)
(26)			MCWB	4.8	4.5	8.7	12.5	17.1	20.1	21.1	20.3	18.5	14.8	10.3	7.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	13.6	15.7	18.3	22.2	24.0	25.5	24.4	23.5	21.3	16.1	14.2	(27)
(28)			MADB	16.2	16.4	21.8	24.7	28.0	29.2	30.9	29.4	28.6	24.5	19.3	16.5	(28)
(29)		2%	WB	11.9	10.9	13.7	16.4	20.5	23.0	24.1	23.2	21.9	19.0	14.1	11.8	(29)
(30)			MADB	13.3	12.9	18.2	21.9	25.8	28.3	29.1	27.7	26.4	22.2	16.9	13.3	(30)
(31)		5%	WB	8.3	7.8	11.8	15.1	19.4	22.0	23.2	22.3	20.7	16.8	12.4	9.3	(31)
(32)			MADB	10.0	10.3	15.4	19.9	24.2	26.9	27.8	26.5	24.6	20.0	15.2	11.2	(32)
(33)		10%	WB	5.0	4.8	9.3	13.6	18.0	21.0	22.3	21.5	19.6	15.1	10.6	6.6	(33)
(34)			MADB	7.1	7.9	13.6	18.2	22.3	25.1	26.3	25.3	22.9	18.6	13.9	8.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	9.0	10.0	11.9	11.7	11.1	11.1	11.0	10.4	8.8	7.4	(35)	
(36)			MCDBR	10.3	13.1	14.3	15.4	14.2	12.8	12.5	12.7	13.2	13.2	12.1	11.4	(36)
(37)			MCWBR	8.8	10.2	8.9	8.7	7.4	6.4	5.8	5.9	6.7	7.9	8.6	9.5	(37)
(38)		5% WB	MCDBR	9.8	11.9	12.2	12.3	11.9	11.1	10.3	10.2	10.8	10.9	10.4	10.1	(38)
(39)			MCWBR	9.1	10.3	8.8	8.1	6.7	6.0	5.4	5.6	6.0	7.5	8.4	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.328	0.354	0.378	0.416	0.451	0.454	0.447	0.383	0.362	0.318	0.300	(40)	
(41)		taud	2.444	2.352	2.364	2.330	2.236	2.172	2.181	2.190	2.346	2.406	2.511	2.503	(41)	
(42)		Ebn at Noon	858	881	896	894	863	830	823	818	853	831	834	833	(42)	
(43)		Edh at Noon	78	100	110	122	138	147	145	139	111	94	74	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.47	2.31	3.28	4.48	5.23	5.69	5.74	5.17	4.21	2.77	1.86	1.21	(44)	
(45)		RadStd	0.14	0.32	0.33	0.45	0.46	0.48	0.40	0.30	0.43	0.28	0.25	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page